American Computer Science League

2024-2025 ● Contest 2: Short Problems Solutions ● Senior Division

1. Prefix/Infix/Postfix Notation

a b / c / d 2 ^ e / − +
= (a b /) c / (d 2 ^) e / − +
= ((/ a b) c /) ((^ d 2) e /) − +
= ((/ / a b c) (/ ^ d 2 e) −) +
= − / / a b c / ^ d 2 e

D. − / / a b c / ^ d 2 e

2. Prefix/Infix/Postfix Notation

4 2 * 4 3 + # 2 / 5 9 − 1 # +
= (4 2 *) (4 3 +) # 2 / (5 9 −) 1 # +
= (8 7 #) 2 / ((-4) 1 #) +
= (8 2 /) 1 +
= 4 1 +
= 5

C. 5

3. Bit-String Flicking

(NOT 11010) OR (10111 AND 10001) XOR 10101
= 00101 OR (10001 XOR 10101)
= 00101 OR 00100
= 00101

A. 00101

4. Bit-String Flicking

Let X = abcde

LHS = (LCIRC-1 (10110 OR 01110)) AND X
= (LCIRC-1 11110) AND abcde
= 11101 AND abcde
= abc0e
RHS = 10001

LHS = RHS ⇒abc0e = 10001 ⇒ a = 1, b = 0, c = 0, d = *, e = 1
Therefore X = 100*1

C. 100*1

5. LISP (ADD (EXP 2 4) (SUB 8 3) (MULT 2 1 3 4)) = (ADD (2^4) (8 − 3) (2 * 1 * 3 * 4)) = (ADD 16 5 24) = 45	E. 45
6. LISP (CDR (CAR (CDR (CDR '((a (b c)) d (e (f g) h) j))))) = (CD R(CAR (CDR '(d (e (f g) h) j)))) = (CDR (CAR '((e (f g) h) j))) = (CDR (e (f g) h)) = ((f g) h)	A. ((f g) h)